

Chapter 11: Three Dimensional Geometry

Q. No.	Question	Marks
1	A bullet shot from the gun travels a straight line path which makes angles 90° , 60° and 30° with the positive direction of x-axis, y-axis and z-axis respectively. Its direction cosines are (a) $1, \frac{\sqrt{3}}{2}, \frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$ (c) $0, \frac{1}{2}, \frac{\sqrt{3}}{2}$ (d) none of these	1
2	The direction cosines of an electricity straight wire with direction ratios 2, -3, 4 are (a) 2, -3, 4 (b) 4, -6, 8 (c) $\frac{2}{\sqrt{29}}, \frac{-3}{\sqrt{29}}, \frac{4}{\sqrt{29}}$ (d) $\frac{4}{\sqrt{29}}, \frac{-6}{\sqrt{29}}, \frac{8}{\sqrt{29}}$	1
3	Three stars in sky are positioned at A(2, -4, 6), B(4, 6, -8) and C(6, 16, -22) with respect to a common reference point O(0, 0, 0). A student is confused whether those three stars are in same line or not. He asks his teacher to help him to solve this problem. Help him to answer this question. (a) Three stars are collinear (b) Three stars are not in a same line (c) AB is perpendicular to BC (d) none of these	1
4	Find the direction ratios of a ray of light passing through the points (1, 2, 3) and (-1, -3, 5). (a) -2, 5, 2 (b) -2, -5, 2 (c) -2, -5, 8 (d) 2, -5, 8	1
5	What are direction ratios of the line $\vec{r} = (3\hat{i} + 4\hat{j} - 5\hat{k}) + m(7\hat{j} + 3\hat{k})$? (a) 3, 4, -5 (b) -3, -4, 5 (c) 3, 11, -2 (d) 0, 7, 3	1
6	What are the direction cosines of the line having direction ratios 0, -3, 4? (a) 0, -3, 4 (b) 0, -8, 10 (c) 0, 3/5, 4/5 (d) 0, -3/5, 4/5	1
7	Find the Cartesian equation of a line parallel to y-axis and passing through the point (1, -2, 7) (a) $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-7}{7}$ (b) $\frac{x-1}{1} = \frac{y+2}{0} = \frac{z-7}{1}$ (c) $\frac{x+1}{1} = \frac{y-2}{-2} = \frac{z+7}{7}$ (d) $\frac{x-1}{0} = \frac{y+2}{1} = \frac{z-7}{0}$	1
8	Write down the vector form of the following equation of line $\frac{x-6}{2} = \frac{y-4}{1} = \frac{z-1}{-3}$ (a) $\vec{r} = (6\hat{i} + 4\hat{j} + 1\hat{k}) + \alpha(2\hat{i} + \hat{j} - 3\hat{k})$ (b) $\vec{r} = (2\hat{i} + \hat{j} - 3\hat{k}) + \alpha(6\hat{i} + 4\hat{j} + 1\hat{k})$ (c) $\vec{r} = (-2\hat{i} - \hat{j} + 3\hat{k}) + \alpha(6\hat{i} + 4\hat{j} + 1\hat{k})$ (d) $\vec{r} = (-6\hat{i} - 4\hat{j} - 1\hat{k}) + \alpha(2\hat{i} + \hat{j} - 3\hat{k})$	1
9	Two lines with direction ratios a, b, c and p, q, r respectively are said to be if $ap + bq + cr = 0$. (a) Parallel (b) Perpendicular (c) Coincident (d) Skew	1
10	For what value of p, given two lines are parallel? $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-7}{7}$ and $\frac{x-8}{2} = \frac{y-2}{p} = \frac{z+2}{14}$ (a) p = -2 (b) p = 4 (c) p = -4 (d) can't be determined	1
11	If a line has direction ratios 2, -1, -2, determine its direction cosines: (a). $\frac{1}{3}, \frac{2}{3}, -\frac{1}{3}$ (b). $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$ (c). $-\frac{2}{3}, \frac{1}{3}, \frac{2}{3}$ (d). None of the above	1
12	The direction ratios of the line $6x - 2 = 3y + 1 = 2z - 2$ are: (a) 6, 3, 2 (b) 1, 1, 2 (c) 1, 2, 3 (d) 1, 3, 2	1
13	What are the direction cosines of x axis? (a) (1,0,0) (b) (0,1,1) (c) (0,0,1) (d) (0,1,1)	1
14	What is the angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$? (a) 0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$	1
15	Assertion (A): The vector form of the line through the point (5, 2, -4) which is parallel to the vector $2\hat{i} + \hat{j} - 6\hat{k}$ is $\vec{r} = 2\hat{i} + \hat{j} - 6\hat{k} + s(5\hat{i} + 2\hat{j} - 4\hat{k})$	1

	Reason (R) : Vector equation of a line passing through the given point with position vector $\vec{a}$ and parallel to the given vector $\vec{b}$ is $\vec{r} = \vec{a} + s \vec{b}$ (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	
16	Assertion (A): Skew lines are non-intersecting and non-parallel lines. Reason (R) : They exist in 3D space only. (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	1
17	Assertion (A) : The angle between the diagonals of a cube is $\frac{1}{3}$ Reason (R) : The D.R.s of the diagonals of a cube are proportional to a, a, a and -a, a, a (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	1
18	Assertion (A) : The image of (0, 2, 0) in x-axis is (0, -2, 0) Reason (R) : x-axis is perpendicular to y-axis and with reference to (0, 2, 0), (0, 0, 0) is foot of the perpendicular on x-axis. (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	1
19	Assertion (A) : If a line makes an angle of $\frac{\pi}{4}$ with each of y and z axis then it makes a right angle with x-axis Reason (R) : The sum of the angles made by a line with the coordinate axes is π (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	1
20	Assertion (A) : The acute angle between the line $\vec{r} = (\hat{i} + \hat{j} + 2\hat{k}) + \beta (\hat{i} - \hat{j})$ and x-axis is $\frac{\pi}{4}$ Reason (R) : If θ is the acute angle between $\vec{r} = \vec{a}_1 + \beta \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \beta \vec{b}_2$, then $\cos \theta = \frac{ \vec{b}_1 \cdot \vec{b}_2 }{\ \vec{b}_1\ \ \vec{b}_2\ }$ (a) Both A and R are true and R is correct explanation of A (b) Both A and R are true but R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	1
21	Check whether the given two lines are coincident, skew, parallel or perpendicular? $\vec{r} = (6\hat{i} + 4\hat{j} + 1\hat{k}) + \beta (2\hat{i} + \hat{j} - 3\hat{k})$ $\vec{r} = (-2\hat{i} - \hat{j} + 3\hat{k}) + \alpha (6\hat{i} + 3\hat{j} - 9\hat{k})$	2
22	Find the angle between the pair of lines: $\vec{r} = (6\hat{i} + 4\hat{j} - 8\hat{k}) + \gamma (2\hat{i} + 4\hat{j} + 4\hat{k})$ and $\vec{r} = (10\hat{i} - 4\hat{j}) + \delta (6\hat{i} + 4\hat{j} + 12\hat{k})$	2
23	Find the direction cosines of the sides of the triangle with vertices A (1, 3, 5), B (2, 5, 7) and C (-1, -4, 3)	2
24	Show that the line passing through two points (2, 3, 5) and (5, 6, 8) is parallel to the line through the points (1, 6, -5) and (4, 9, -2).	2
25	Find the angle between the pair of lines:	2

	$\vec{r} = (8\hat{i} + 4\hat{j} + 9\hat{k}) + \beta (2\hat{i} + \hat{j} - 3\hat{k})$ $\vec{r} = (7\hat{i} - 3\hat{j} + 1\hat{k}) + \alpha (6\hat{i} + 3\hat{j} - 9\hat{k})$	
26	Find the value of 'm' so that the given lines are perpendicular. $\frac{-x+1}{1} = \frac{y+2}{2} = \frac{z-7}{4} \quad \text{and} \quad \frac{x-8}{2} = \frac{y-2}{m} = \frac{z+2}{6}$	2
27	Find the value of λ , so that the following lines are perpendicular to each other $\frac{x-5}{5\lambda+2} = \frac{2-y}{5} = \frac{1-z}{-1} \quad \text{and} \quad \frac{x}{1} = \frac{2y+1}{4\lambda} = \frac{1-z}{-3}$	2
28	Find the shortest distance between the following lines: $\vec{r} = (2\hat{i} + 4\hat{j} - 8\hat{k}) + \beta (2\hat{i} + 3\hat{j} + 6\hat{k})$ $\vec{r} = (\hat{i} - 2\hat{j} - 4\hat{k}) + \alpha (4\hat{i} + 6\hat{j} + 12\hat{k})$	3
29	Find the equation of a line parallel to $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ and passing through $2\hat{i} + 4\hat{j} + 5\hat{k}$. Also find the S.D. between these lines.	3
30	Find the shortest distance between the lines $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$ and $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$	3
31	Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$	3
32	Find the equation of the line passing through (1, -1, 1) and perpendicular to the lines joining the points (4, 3, 2), (1, -1, 0) and (1, 2, -1), (2, 2, 1).	4
33	Find the point on the line $\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2}$ at a distance 5 units from the point P(1, 3, 3)	4
34	Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Find their point of intersection.	4
35	Find the image of the point (1, 6, 3) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$.	4
36	Find the coordinates of the foot of the perpendicular drawn from the point A(1, 8, 4) to the line joining the points B(0, -1, 3) and C(2, -3, -1).	4
37	The equation of motion of a missile are $x = 2t$, $y = 3t$, $z = t$, where the time 't' is given in seconds and distance is measured in kilometers. Based on it, answer the following question; (i) What is the path of the missile? (a) Straight line (b) Parabola (c) Circle (d) Ellipse (ii) Which of the following points lie on the path of missile? (a) (1, 2, 3) (b) (2, 3, 1) (c) (4, 1, -2) (d) (1, -2, 3) (iii) At what distance will the missile be in 10 seconds from the starting point (0, 0, 0)? (a) $10\sqrt{14}$ km (b) $20\sqrt{14}$ km (c) $10\sqrt{7}$ km (d) $20\sqrt{14}$ km (iv) The position of missile at a certain instant of time is (2, -8, 15) then what will be height of the missile from the ground if ground is considered as xy-plane? (a) 2 km (b) 8 km (c) 15 km (d) 7 km	4
38	In a class, teacher asks students what they know about space or three dimensional system. He asks students some basic questions. Help students to answer the following; (i) What is the equation of x-axis in space? (a) $x = 0, y = 0$ (b) $y = 0, z = 0$ (c) $x = 0$ (d) none of these (ii) What are direction ratios of y-axis? (a) 0,0,1 (b) 1,0,0 (c) 0,1,0	4

	(d) 1,0,1 (iii) Direction cosines of a line are $\langle m, m, m \rangle$, then (a) $m > 0$ (b) $m < 0$ (c) $m < 1$ (d) $m = \frac{1}{\sqrt{3}}$ or $\frac{-1}{\sqrt{3}}$ (iv) Which of the following statement is correct? (a) Direction ratios of a line are equal to its direction cosines. (b) Direction ratios of two perpendicular lines are proportional. (c) Direction ratios of two parallel lines are proportional. (d) All of these are correct.	
39	Find the coordinates of the image of the point (2, 3, 4) with respect to the line $\vec{r} = (2\hat{j} + 4\hat{k}) + \gamma (2\hat{i} + 4\hat{j} + 1\hat{k})$; where γ is a scalar. Also, find the distance of the image from the origin.	4
40	An aeroplane is flying along the line $\vec{r} = \alpha (2\hat{i} + 3\hat{j} + 4\hat{k})$; where α is a scalar and another aeroplane is flying along the line $\vec{r} = (\hat{i} + \hat{j}) + \gamma (3\hat{j} + 2\hat{k})$; where γ is a scalar. At what points on the line should they reach, so that the distance between them is shortest. Find the shortest possible distance between them.	4